**Design to Cost**
PART COST ESTIMATOR

2026-10-26

RAW-MATERIAL COST SHEET

# EX26-0001 · L mounting bracket

Rev B · Galvanised steel GI · IN · costing month 2026-10 · version 2

**Uses example rates – not for negotiation**
Replace the EXAMPLE rates marked below on the Rates page first.

## Result

| | INR | INR |
|---|---|---|
| **Net RM cost per part** | **15.8374** | **15.8374** |

## Steps and formulas

| Step | Formula | Value | Unit |
|---|---|---|---|
| Net weight | volume * density | 199.8304 | g |
| Blanks per sheet | best straight grid of blank_length_mm × blank_width_mm on the usable sheet (sheet_length_mm – 2 × edge_mm) × (sheet_width_mm – 2 × edge_mm), gap_mm between blanks, both orientations | 189 | pcs |
| Sheet weight (nesting information) | sheet_length_mm * sheet_width_mm * thickness_mm / 1000 * density_g_cm3 | 49,062 | g |
| Blank weight | blank_length_mm * blank_width_mm * thickness_mm / 1000 * density_g_cm3 | 205.1676 | g |
| Gross weight (calculated blank + gap) | (blank_length_mm + gap_mm) * (blank_width_mm + gap_mm) * thickness_mm / 1000 * density_g_cm3 | 234.346 | g |
| Skeleton & punched-out scrap | gross_g - net_weight_g | 34.5157 | g |
| Material used per part | gross_g | 234.346 | g |
| Material yield | 100 * net_weight_g / input_g | 85.2715 | % |
| Sheet utilisation | 100 * net_weight_g * blanks_per_sheet / sheet_weight_g | 76.9792 | % |
| RM cost (material used × rate) | input_g / 1000 * material_rate | 16.8729 | INR |
| Scrap credit (–) | 0 - returns_g / 1000 * scrap_rate | -1.0355 | INR |
| Net RM cost per part | rm_cost + scrap_credit | 15.8374 | INR |
| Net RM cost per part in INR | value * fx | 15.8374 | INR |

## Inputs and where each comes from

| Input | Value | Unit | Comes from |
|---|---|---|---|
| Volume | 25.4561 | cm³ | 3D file example_bracket_revB.step · rev B · Measured from the STEP file with OpenCascade (CadQuery) |
| Surface area | 272.4334 | cm² | 3D file example_bracket_revB.step · rev B · Measured from the STEP file with OpenCascade (CadQuery) |
| Projected area (along Z) | 97.734 | cm² | 3D file example_bracket_revB.step · rev B · Measured from the STEP file with OpenCascade (CadQuery) |
| Density of Galvanised steel GI | 7.85 | g/cm³ | Master data → Materials · Galvanised steel GI |
| Sheet thickness | 2 | mm | 3D file example_bracket_revB.step · rev B · Most common wall (sheet thickness) found in the 3D model |
| Flat blank length | 326.7 | mm | 3D file example_bracket_revB.step · rev B · Developed blank from the 3D model: Developed straight strip + bending stage (100 % material yield) |
| Flat blank width | 40 | mm | 3D file example_bracket_revB.step · rev B · Developed blank from the 3D model: Developed straight strip + bending stage (100 % material yield) |
| Sheet length | 2,500 | mm | Master data → Parameters · sheet_metal / sheet_length_mm |
| Sheet width | 1,250 | mm | Master data → Parameters · sheet_metal / sheet_width_mm |
| Gap between blanks | 5 | mm | Master data → Parameters · sheet_metal / gap_mm |
| Sheet edge trim | 5 | mm | Master data → Parameters · sheet_metal / edge_mm |
| Material rate | 72 | INR/kg | Rate library #585 · material / Galvanised steel GI · IN · 2026-10 · approved · EXAMPLE |
| Scrap value | 30 | INR/kg | Rate library #586 · material_scrap / Galvanised steel GI · IN · 2026-10 · approved · EXAMPLE |
| INR per INR | 1 | INR/INR | INR is the base currency. |

## Rate versions used (the report can be reproduced with these)

| Id | Rate | Country | Month | Value | Unit | Status | Approved by | |
|---|---|---|---|---|---|---|---|---|
| 585 | material / Galvanised steel GI | IN | 2026-10 | 72 | INR/kg | approved | Example seed | EXAMPLE |
| 586 | material_scrap / Galvanised steel GI | IN | 2026-10 | 30 | INR/kg | approved | Example seed | EXAMPLE |

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